

Fine-Grained Emotion Structure in the Brain-Predictable Subspace of a Self-Supervised Video Model

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Abstract

Do the brain and self-supervised video models share a common structure for representing emotion? A central debate in affective neuroscience concerns whether emotions are represented along affective dimensions such as valence and arousal, or as fine-grained emotion categories. Prior work showed that neural responses to emotional videos are better explained by fine-grained emotion categories than by affective dimensions. Here, we ask whether this categorical structure extends to the subspace shared between the human brain and a video foundation model trained without emotion labels. By aligning neural representations from a brain foundation model (Brain-JEPA) with embeddings from a self-supervised video foundation model (V-JEPA2), we identified a compact brain-predictable subspace restricted to only three of the leading principal components of a 1,408-dimensional embedding. This shared neural-computational subspace preferentially encodes discrete emotion categories rather than affective dimensions, with categorical dominance exceeding that of the full video model space, a pattern stable across all individual subjects. These findings suggest that self-supervised learning from visual experience spontaneously produces a low-dimensional, categorically organized affective subspace that aligns with human neural emotion representations.

Keywords: emotion representation, fMRI, video foundation model, self-supervised learning, brain-predictable subspace

Introduction

A central question in affective neuroscience concerns how the brain organizes emotional experience, whether along continuous affective dimensions such as valence and arousal (Russell, 1980), or as a high-dimensional semantic space of fine-grained emotion categories (Cowen and Keltner, 2017). To approach this question, Horikawa et al. (2020) collected whole-brain fMRI responses while subjects viewed over 2,000 emotionally evocative videos, and demonstrated that neural responses are better predicted by fine-grained emotion categories than by affective dimensions, with emotion-related representations distributed across transmodal brain regions. These findings suggest that the brain represents emotion in a high-dimensional, categorically organized geometry. Recent work at the intersection of AI and affective neuroscience has shown that artificial neural networks can successfully predict human

emotional responses to visual stimuli, suggesting that emotion schemas are deeply embedded within both biological and artificial visual systems (Kragel et al., 2019; Conwell et al., 2025). However, these studies have primarily relied on supervised models trained with explicit conceptual or object categories. In parallel, self-supervised video foundation models learn structured visual representations from large-scale video data without any semantic or affective supervision. Here we examine the affective geometry of the subspace shared between such models and the brain, by aligning neural representations from Brain-JEPA (Dong et al., 2024) with embeddings from V-JEPA2 (Assran et al., 2025), a self-supervised video foundation model trained on over one million hours of video. We find that the brain-predictable subspace of V-JEPA2 is compact and preferentially encodes categorical rather than dimensional emotion structure.

Methods

We used the fMRI dataset from Horikawa et al. (2020), in which five healthy adult subjects each viewed 2,196 emotionally evocative short videos. Emotion ratings were obtained from independent crowd workers, yielding continuous scores for 34 emotion categories and 14 affective dimension ratings (Cowen and Keltner, 2017). Neural representations were extracted per subject by passing preprocessed fMRI through Brain-JEPA (Dong et al., 2024), a self-supervised fMRI foundation model pretrained on UK Biobank data, without task-specific fine-tuning. Subject-level embeddings were averaged across subjects for group-level analyses. Video representations were obtained from V-JEPA2 (Assran et al., 2025) by uniformly sampling 16 frames per video and mean-pooling the final hidden state across tokens, yielding a (2196, 1408) embedding. To identify which dimensions of V-JEPA2 covary with brain activity, we first decomposed the video embeddings into 100 principal components (PCs) to capture the majority of variance, and predicted each PC from the group-mean Brain-JEPA representation via 5-fold cross-validated ridge regression. PCs surviving permutation-based FDR correction ($q < 0.05$, $n = 1,000$ permutations) were defined as brain-predictable. We then examined the affective structure of this brain-predictable subspace. Using the same regression pipeline, we decoded 34 emo-

tion category scores and, among the 14 affective dimensions, focused on valence and arousal (V-A) as the two canonical axes of the dimensional model of emotion. Subject-level stability was assessed by repeating the brain-predictability and decoding analyses for each subject individually. To characterize the affective geometry of the brain-predictable subspace, we computed the ratio of mean category R^2 to mean V-A R^2 (category/V-A ratio). We computed the same ratio for the full video model space (all 100 PCs) as a baseline comparison. To control for low-level visual and semantic confounds, VGG19-based vision features (1,000-dim) and human-annotated semantic features (73-dim) were regressed out from both representations and targets, and partial R^2 was recomputed.

Results

Of 100 principal components, only three survived permutation-based FDR correction (PC1: $R^2 = 0.373$, PC2: $R^2 = 0.075$, PC3: $R^2 = 0.088$), while all remaining components fell near zero. This indicates a highly compressed shared space, anchored by PC1 which remained stable across all five individual subjects. These brain-predictable components showed substantially stronger maximum absolute Spearman correlations with emotion categories than brain-unpredictable components (mean max $|r| = 0.33$ vs. 0.09, FDR-corrected), confirming that the shared neural-computational subspace is affectively organized (Figure 1).

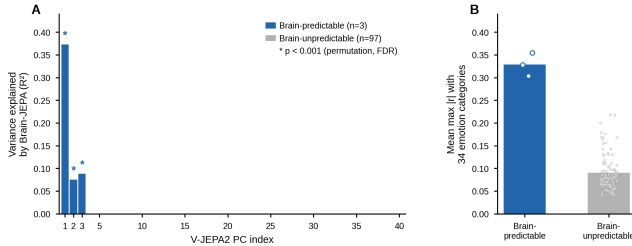


Figure 1: Brain-predictable subspace of V-JEPA2. (A) R^2 for predicting each V-JEPA2 PC from Brain-JEPA. Only PC1–3 survived permutation-based FDR correction. (B) Mean max $|r|$ with emotion categories for brain-predictable vs. brain-unpredictable PCs (0.33 vs. 0.09).

Decoding from this brain-predictable subspace revealed stronger prediction of emotion categories than valence and arousal (mean $R^2 = 0.055$ vs. 0.038, category/V-A ratio = 1.44). This categorical dominance exceeded that of the full V-JEPA2 space (ratio = 1.26), indicating that the dimensions selectively shared with the brain are more categorically organized than V-JEPA2’s full representational space. This pattern was stable across all five subjects. After regressing out low-

level vision and semantic features, absolute predictive values decreased substantially, though the categorical organization of the shared subspace was attenuated but preserved (Figure 2).

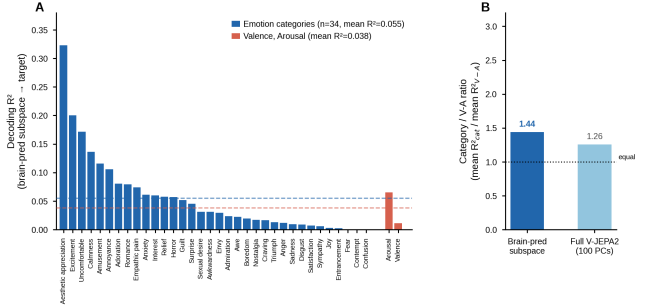


Figure 2: Categorical emotion structure of the brain-predictable subspace. (A) Emotion decoding R^2 from the brain-predictable subspace across 34 categories. Valence and Arousal shown separately. (B) category/V-A ratio for the brain-predictable subspace (1.44) vs. the full V-JEPA2 space (1.26). Ratio > 1 in all five subjects.

Discussion

Our findings reveal that the representational space shared between a self-supervised video foundation model and the human brain is remarkably compact, confined to only three PCs within the top 100 components of a 1,408-dimensional embedding. Despite this sparsity, the shared subspace organizes emotion along categorical rather than dimensional lines, mirroring the categorical dominance observed by Horikawa et al. (2020), and supporting theoretical arguments that emotion schemas are embedded within the statistical regularities of the visual environment (Kragel et al., 2019; Conwell et al., 2025). The shared signal, however, substantially overlaps with low-level visual and semantic structure, suggesting the subspace reflects affective organization partly grounded in visual content rather than purely abstract emotional representations. This has practical relevance for brain-inspired affective AI. Recent work shows that fine-tuning foundation models on neural responses, termed brain-tuning, improves brain alignment and downstream task performance (Moussa et al., 2025). The compact categorical subspace identified here provides a principled target for such approaches in the affective domain. Future work should examine whether this structure generalizes across different video models and neuroimaging modalities.

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